In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
sales = pd.read_csv("C:/Users/Ganesh/Downloads/SuperMarket Analysis.csv")
```

In [3]:
```
sales.head()
```

Out[3]:

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	
0	750-67-8428	Alex	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	54
1	226-31-3081	Giza	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200	8
2	631-41-3108	Alex	Yangon	Normal	Female	Home and lifestyle	46.33	7	16.2155	34
3	123-19-1176	Alex	Yangon	Member	Female	Health and beauty	58.22	8	23.2880	48
4	373-73-7910	Alex	Yangon	Member	Female	Sports and travel	86.31	7	30.2085	63

In [4]:
```
sales.tail()
```

Out[4]:

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%
995	233-67-5758	Giza	Naypyitaw	Normal	Male	Health and beauty	40.35	1	2.0175
996	303-96-2227	Cairo	Mandalay	Normal	Female	Home and lifestyle	97.38	10	48.6900
997	727-02-1313	Alex	Yangon	Member	Male	Food and beverages	31.84	1	1.5920
998	347-56-2442	Alex	Yangon	Normal	Male	Home and lifestyle	65.82	1	3.2910
999	849-09-3807	Alex	Yangon	Member	Female	Fashion accessories	88.34	7	30.9190

In [5]:
```
sales.shape
```

Out[5]:
```
(1000, 17)
```

In [6]:
```
sales.isnull().sum()
```

Out[6]:
```
Invoice ID                0
Branch                    0
City                      0
Customer type             0
Gender                    0
Product line              0
Unit price                0
Quantity                  0
Tax 5%                    0
Sales                     0
Date                      0
Time                      0
Payment                   0
cogs                      0
gross margin percentage   0
gross income              0
Rating                    0
dtype: int64
```

In [7]:
```
sales.count()
```

Out[7]:
```
Invoice ID                 1000
Branch                     1000
City                       1000
Customer type              1000
Gender                     1000
Product line               1000
Unit price                 1000
Quantity                   1000
Tax 5%                     1000
Sales                      1000
Date                       1000
Time                       1000
Payment                    1000
cogs                       1000
gross margin percentage    1000
gross income               1000
Rating                     1000
dtype: int64
```

In [8]:
```
sales.sum()
```

Out[8]:
```
Invoice ID                 750-67-8428226-31-3081631-41-3108123-19-117637...
Branch                     AlexGizaAlexAlexAlexGizaAlexGizaAlexCairoCairo...
City                       YangonNaypyitawYangonYangonYangonNaypyitawYang...
Customer type              MemberNormalNormalMemberMemberMemberMemberMemb...
Gender                     FemaleFemaleFemaleFemaleFemaleFemaleFemaleFema...
Product line               Health and beautyElectronic accessoriesHome an...
Unit price                                                          55672.13
Quantity                                                                5510
Tax 5%                                                             15379.369
Sales                                                             322966.749
Date                       01-05-201903-08-201903-03-20191/27/201902-08-2...
Time                       1:08:00 PM10:29:00 AM1:23:00 PM8:33:00 PM10:37...
Payment                    EwalletCashCredit cardEwalletEwalletEwalletEwa...
cogs                                                               307587.38
gross margin percentage                                          4761.904762
gross income                                                       15379.369
Rating                                                                6972.7
dtype: object
```

In [17]:
```
sales['Total'].sum()
```

Out[17]:
```
np.float64(734229.601762)
```

In [18]:
```
sales[['Total','gross income']].sum()
```

Out[18]:
```
Total           734229.601762
gross income     15379.369000
dtype: float64
```

In [25]:
```
sales_mean = sales.mean(numeric_only=True)
print("Mean of numeric columns:")
print(sales_mean)
```

```
Mean of numeric columns:
Unit price                  55.672130
Quantity                     5.510000
Tax 5%                      15.379369
Sales                      322.966749
cogs                       307.587380
gross margin percentage      4.761905
gross income                15.379369
Rating                       6.972700
Total                      734.229602
dtype: float64
```

In [26]:
```
sales['Rating'].mean()
```

Out[26]:
```
np.float64(6.9727)
```

In [30]:
```
numeric_std = sales.select_dtypes(include=['number']).std()
print(numeric_std)
```

```
Unit price                 2.649463e+01
Quantity                   2.923431e+00
Tax 5%                     1.170883e+01
Sales                      2.458853e+02
cogs                       2.341765e+02
gross margin percentage    6.131498e-14
gross income               1.170883e+01
Rating                     1.718580e+00
Total                      5.226213e+02
dtype: float64
```

In [31]:
```
sales['Rating'].std()
```

Out[31]:
```
1.718580294379123
```

In [32]:
```
print('The Mode of : ',sales['Payment'].mode)
```

```
The Mode of :  <bound method Series.mode of 0          Ewallet
1             Cash
2      Credit card
3          Ewallet
4          Ewallet
          ...
995        Ewallet
996        Ewallet
997           Cash
998           Cash
999           Cash
Name: Payment, Length: 1000, dtype: object>
```

In [33]:
```
sales['Payment'].mode()
```

Out[33]:
```
0    Ewallet
Name: Payment, dtype: object
```

In [34]:
```
sales['Quantity'].min()
```

Out[34]: 1

In [35]:
```
sales['Quantity'].max()
```

Out[35]: 10

In [36]:
```
sales['Total'].cumsum().head()
```

Out[36]:
```
0    1219.636405
1    1418.538310
2    2181.296714
3    3262.062619
4    4664.400024
Name: Total, dtype: float64
```

In [37]:
```
sales['Total'].head()
```

Out[37]:
```
0    1219.636405
1     198.901905
2     762.758405
3    1080.765905
4    1402.337405
Name: Total, dtype: float64
```

In [40]:
```
# Add a semicolon or newline between the two statements
median = sales['Quantity'].median()
median

# Alternatively, you could use a semicolon
# median = sales['Quantity'].median(); median
```

Out[40]: 5.0

In [41]:
```
print('median quantity: ' + str(median))
```

```
median quantity: 5.0
```

In [42]:
```
var = sales['Quantity'].var()
var
```

Out[42]: 8.546446446446451

In [43]:
```
print('var of quantity: ' + str(var))
```

```
var of quantity: 8.546446446446451
```

In [45]:
```
groupby_sum = sales.groupby(['City']).sum()
groupby_sum
```

Out[45]:

City	Invoice ID	Branch	
Mandalay	692-92-5582351-62-0822529-56-3974299-46-180531...	CairoCairoCairoCairoCairoCairoCairoCairoC...	MemberMemberMembe
Naypyitaw	226-31-3081699-14-3026315-22-5665300-71-460518...	GizaGizaGizaGizaGizaGizaGizaGizaGizaGizaGi...	NormalMemberMembe
Yangon	750-67-8428631-41-3108123-19-1176373-73-791035...	AlexAlexAlexAlexAlexAlexAlexAlexAlexAlexAl...	MemberNormalMembe

In [46]:
```
groupby_count = sales.groupby(['City']).count()
groupby_count
```

Out[46]:

City	Invoice ID	Branch	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Sales	Dat
Mandalay	332	332	332	332	332	332	332	332	332	33
Naypyitaw	328	328	328	328	328	328	328	328	328	32
Yangon	340	340	340	340	340	340	340	340	340	34

In [47]:
```
groupby_sum1 = sales.groupby(['Payment']).sum()
groupby_sum1
```

Out[47]:

Payment	Invoice ID	Branch
Cash	226-31-3081529-56-3974829-34-3910299-46-180564...	GizaCairoAlexCairoCairoCairoAlexCairoCairoCair...	NaypyitawMandalayYango
Credit card	631-41-3108665-32-9167692-92-5582656-95-934976...	AlexAlexCairoAlexAlexAlexCairoAlexAlexCairoAle...	YangonYangonMandalayYar
Ewallet	750-67-8428123-19-1176373-73-7910699-14-302635...	AlexAlexAlexGizaAlexGizaCairoAlexAlexCairoGiza...	YangonYangonYangonNayp

In [48]:
```
groupby_count1 = sales.groupby(['Payment']).count()
groupby_count1
```

Out[48]:

Payment	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Sales
Cash	344	344	344	344	344	344	344	344	344	344
Credit card	311	311	311	311	311	311	311	311	311	311
Ewallet	345	345	345	345	345	345	345	345	345	345

In [49]:
```
print('sum of values, grouped by the payment: ' + str(groupby_sum1))
```

```
sum of values, grouped by the payment:
Invoice ID  \
Payment
Cash         226-31-3081529-56-3974829-34-3910299-46-180564...
Credit card  631-41-3108665-32-9167692-92-5582656-95-934976...
Ewallet      750-67-8428123-19-1176373-73-7910699-14-302635...

                                                        Branch  \
Payment
Cash         GizaCairoAlexCairoCairoCairoAlexCairoCairoCair...
Credit card  AlexAlexCairoAlexAlexAlexCairoAlexAlexCairoAle...
Ewallet      AlexAlexAlexGizaAlexGizaCairoAlexAlexCairoGiza...

                                                          City  \
Payment
Cash         NaypyitawMandalayYangonMandalayMandalayMandala...
Credit card  YangonYangonMandalayYangonYangonYangonMandalay...
Ewallet      YangonYangonYangonNaypyitawYangonNaypyitawMand...

                                                 Customer type  \
Payment
Cash         NormalMemberMemberMemberMemberMemberMemberMemb...
Credit card  NormalMemberMemberMemberMemberMemberMemberMemb...
Ewallet      MemberMemberMemberMemberMemberMemberMemberMemb...

                                                        Gender  \
Payment
Cash         FemaleFemaleFemaleFemaleFemaleFemaleFemaleFema...
Credit card  FemaleFemaleFemaleFemaleFemaleFemaleFemaleFema...
Ewallet      FemaleFemaleFemaleFemaleFemaleFemaleFemaleFema...

                                                  Product line  Unit price  \
Payment
Cash         Electronic accessoriesElectronic accessoriesHe...    19525.09
Credit card  Home and lifestyleHealth and beautyFood and be...    16916.68
Ewallet      Health and beautyHealth and beautySports and t...    19230.36

             Quantity    Tax 5%       Sales  \
Payment
Cash             1896  5343.170  112206.570
Credit card      1722  4798.432  100767.072
Ewallet          1892  5237.767  109993.107

                                                          Date  \
Payment
Cash         03-08-201903-09-20193/29/20191/15/201902-08-20...
Credit card  03-03-201901-10-20192/20/201903-11-201901-01-2...
Ewallet      01-05-20191/27/201902-08-20193/25/20192/25/201...

                                                          Time       cogs  \
Payment
Cash         10:29:00 AM5:03:00 PM7:21:00 PM4:19:00 PM3:31:...  106863.40
Credit card  1:23:00 PM5:15:00 PM1:27:00 PM11:03:00 AM10:39...   95968.64
Ewallet      1:08:00 PM8:33:00 PM10:37:00 AM6:30:00 PM2:36:...  104755.34

             gross margin percentage  gross income  Rating          Total
```

```
Payment
Cash                    1638.095238      5343.170  2397.7  255213.195238
Credit card             1480.952381      4798.432  2178.0  228630.208381
Ewallet                 1642.857143      5237.767  2397.0  250386.198143
```

In [50]:
```
print('count of values, grouped by the payment: ' + str(groupby_count1))
```

```
count of values, grouped by the payment:              Invoice ID  Branch  City  Cust
omer type  Gender  Product line  \
Payment
Cash                344     344   344            344     344           344
Credit card         311     311   311            311     311           311
Ewallet             345     345   345            345     345           345

             Unit price  Quantity  Tax 5%  Sales  Date  Time  cogs  \
Payment
Cash                344       344     344    344   344   344   344
Credit card         311       311     311    311   311   311   311
Ewallet             345       345     345    345   345   345   345

             gross margin percentage  gross income  Rating  Total
Payment
Cash                             344           344     344    344
Credit card                      311           311     311    311
Ewallet                          345           345     345    345
```

In [51]:
```
sales.describe()
```

Out[51]:

	Unit price	Quantity	Tax 5%	Sales	cogs	gross margin percentage	
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.00000	1.000000e+03	1000
mean	55.672130	5.510000	15.379369	322.966749	307.58738	4.761905e+00	15
std	26.494628	2.923431	11.708825	245.885335	234.17651	6.131498e-14	11
min	10.080000	1.000000	0.508500	10.678500	10.17000	4.761905e+00	0
25%	32.875000	3.000000	5.924875	124.422375	118.49750	4.761905e+00	5
50%	55.230000	5.000000	12.088000	253.848000	241.76000	4.761905e+00	12
75%	77.935000	8.000000	22.445250	471.350250	448.90500	4.761905e+00	22
max	99.960000	10.000000	49.650000	1042.650000	993.00000	4.761905e+00	49

In [52]:
```
sales.describe(include='object')
```

Out[52]:

	Invoice ID	Branch	City	Customer type	Gender	Product line	Date	Time	Payment
count	1000	1000	1000	1000	1000	1000	1000	1000	1000
unique	1000	3	3	2	2	6	89	506	3
top	750-67-8428	Alex	Yangon	Member	Female	Fashion accessories	02-07-2019	7:48:00 PM	Ewallet
freq	1	340	340	565	571	178	20	7	345

In [53]:
```
sales.describe(include='all')
```

Out[53]:

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity
count	1000	1000	1000	1000	1000	1000	1000.000000	1000.000000
unique	1000	3	3	2	2	6	NaN	NaN
top	750-67-8428	Alex	Yangon	Member	Female	Fashion accessories	NaN	NaN
freq	1	340	340	565	571	178	NaN	NaN
mean	NaN	NaN	NaN	NaN	NaN	NaN	55.672130	5.510000
std	NaN	NaN	NaN	NaN	NaN	NaN	26.494628	2.923431
min	NaN	NaN	NaN	NaN	NaN	NaN	10.080000	1.000000
25%	NaN	NaN	NaN	NaN	NaN	NaN	32.875000	3.000000
50%	NaN	NaN	NaN	NaN	NaN	NaN	55.230000	5.000000
75%	NaN	NaN	NaN	NaN	NaN	NaN	77.935000	8.000000
max	NaN	NaN	NaN	NaN	NaN	NaN	99.960000	10.000000

In [ ]: